Minimum Gaussian energy for parity requires exponential ReLU width

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Abstract

A Boolean function may have a small neural-network implementation while its minimum-energy implementation requires much greater width. We prove an exact example for parity and one-hidden-layer ReLU networks with a constant input coordinate and independent isotropic Gaussian weights in each layer. For an even parity of degree $r \geq 4$, the smallest Gaussian rate energy compatible with output margin $1/2$ is attained if and only if the hidden width is at least 2^{r-1}. Nevertheless the same parity has a bounded-weight shallow teacher of width $O(r^2)$ and a Boolean circuit of size $O(r)$. The proof first determines the sharp correlation of a single ReLU neuron with parity, then characterizes every equality case by its Fourier coefficients. The result concerns exact energy minimization. It is not an exponential sample-complexity or online-mistake lower bound.

1 The question and the result

For a Gaussian prior on network weights, the leading small-noise probability of a prescribed function is determined by the least quadratic weight energy that implements it. The width needed to *implement* a function need not be the width needed to minimize that energy. This distinction matters when comparing strong-prior Bayesian learners at different widths: adding neurons can change the leading function probabilities even when the target was already realizable.

We give an exact separation for parity. Let $d \geq r \geq 2$, let r be even, and write

$$X = \{-1, 1\}^d, \qquad \chi(x) = \prod_{i=1}^{r} x_i.$$

The learner has one hidden layer and a linear readout:

$$z(x) = \sum_{j=1}^{N} v_j \mathrm{ReLU}(b_j + w_j \cdot x), \qquad \mathrm{ReLU}(t) = \max\{t, 0\}. \tag{1}$$

The bias b_j is the weight of an appended constant input coordinate. For the independent Gaussian prior

$$(b_j, w_{j1}, \ldots, w_{jd})_i \sim N(0, N\sigma^2/d), \qquad v_j \sim N(0, \sigma^2/N),$$

the density is proportional to $\exp[-K_N/(2\sigma^2)]$, where

$$K_N = \frac{d}{N} \sum_{j=1}^{N} (b_j^2 + \|w_j\|_2^2) + N \sum_{j=1}^{N} v_j^2. \tag{2}$$

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All first-layer coordinates, including the constant coordinate, have the same variance. Impose the full-cube margin constraints

$$\chi(x)z(x) \geq \frac{1}{2} \qquad (x \in X). \tag{3}$$

Set

$$\rho_r = \frac{\binom{r-2}{(r-2)/2}}{2^{r-1}\sqrt{r}}, \qquad E_* = \frac{\sqrt{d}}{\rho_r}. \tag{4}$$

Theorem 1. *Every width and every network satisfying (3) obey $K_N \geq E_*$. If $r \geq 4$, equality is attainable exactly at the widths*

$$N \geq 2^{r-1}.$$

For $r = 2$, equality is attainable exactly when $N \geq 4$. At any smaller width where (3) is feasible, its minimum energy is strictly greater than E_*.

The target has a bounded-fan-in Boolean circuit of size $O(r)$ and a one-hidden-layer ReLU teacher of width $O(r^2)$, every weight at most one in absolute value, and output margin one.

The strict-margin cell $\chi(x)z(x) > 1/2$ has the same energy infimum as (3): multiply the readout by $1 + \varepsilon$ and let $\varepsilon \downarrow 0$. Thus the theorem also identifies the width at which that cell reaches its optimal leading Gaussian small-noise rate. The claim is about attaining the *exact* minimum. It gives no positive lower bound, uniform in r, on the approximation gap below the width threshold.

Prior work establishes $\Theta(r)$ variation-norm complexity for parity and shows that multidirectional ReLU representations improve on ridge representations [1]. Our proof uses the augmented Euclidean norm, which charges the bias, and determines the sharp single-neuron constant and the exact support size needed for equality.

2 A sharp single-neuron inequality

For $T \subseteq [d]$, put $\chi_T(x) = \prod_{i \in T} x_i$ and $\widehat{h}(T) = \mathbb{E}[\chi_T(x)h(x)]$, with expectation uniform on the cube.

Lemma 2 (Top Fourier coefficient of a monotone function)**.** *If $h : \{-1, 1\}^m \to [0, 1]$ is coordinatewise nondecreasing and $m \geq 1$, then*

$$\left| \mathbb{E}\left[\left(\prod_{i=1}^{m} x_i \right) h(x) \right] \right| \leq \frac{\binom{m-1}{\lfloor (m-1)/2 \rfloor}}{2^m}.$$

Proof. Average h over coordinate permutations. The objective is unchanged, and the resulting value h_j at inputs with j positive coordinates is nondecreasing in j: a uniform j-subset can be coupled with a uniform $(j+1)$-subset by adding a uniformly chosen absent coordinate. Every sequence $0 \leq h_0 \leq \cdots \leq h_m \leq 1$ is a convex combination of the two constant sequences and the threshold sequences $\mathbf{1}\{j \geq t\}$. For a threshold, the alternating sum is

$$\sum_{j=t}^{m} (-1)^j \binom{m}{j} = (-1)^t \binom{m-1}{t-1} \qquad (1 \leq t \leq m).$$

The largest absolute value is the central binomial coefficient in the statement. The constant sequences have zero top coefficient. □

Lemma 3 (Sharp parity correlation)**.** *For even $r \geq 2$ and every $(b, w) \in \mathbb{R}^{d+1}$,*

$$|\mathbb{E}[\chi(x)\mathrm{ReLU}(b + w \cdot x)]| \leq \frac{\binom{r-2}{(r-2)/2}}{2^{r-1}} \min_{i \leq r} |w_i| \leq \rho_r \sqrt{b^2 + \|w\|_2^2}. \tag{5}$$

For unit augmented norm, equality in the last bound holds exactly at

$$b = 0, \qquad w_i = s_i/\sqrt{r} \ (i \leq r), \qquad w_i = 0 \ (i > r), \qquad s_i \in \{-1, 1\}.$$

Proof. Input sign changes make the relevant w_i nonnegative and change at most the sign of the correlation. Choose one of their smallest values, $a = w_j$. Condition on the irrelevant inputs. Since the derivative of ReLU is the positive-half-line indicator almost everywhere,

$$\mathbb{E}[\chi\mathrm{ReLU}(b + w \cdot x)] = \frac{1}{2} \int_{-a}^{a} \mathbb{E}\left[\left(\prod_{i \leq r,\ i \neq j} x_i\right) \mathbf{1}\left\{b + t + \sum_{i \neq j} w_i x_i > 0\right\}\right] dt.$$

The inner function is monotone in the remaining $r - 1$ relevant bits. Lemma 2 bounds its top coefficient by $\binom{r-2}{(r-2)/2}/2^{r-1}$, proving the first inequality. Next,

$$\min_{i \leq r} |w_i| \leq \frac{\|(w_1, \ldots, w_r)\|_2}{\sqrt{r}} \leq \frac{\sqrt{b^2 + \|w\|_2^2}}{\sqrt{r}}.$$

Equality forces the stated directions. They attain equality: at equal weights and bias zero, the inner threshold in the integral is the same central Hamming threshold for almost every $t \in (-a, a)$. Its top coefficient has constant sign and maximal magnitude. □

For completeness, an odd parity of degree $r \geq 3$ has sharp constant ρ_{r+1} when the bias is charged in the augmented norm. Indeed $\mathrm{ReLU}(t) = (t + |t|)/2$, the affine part has zero parity correlation, and randomizing the sign of the bias converts the absolute-value correlation into an even $(r + 1)$-variable parity correlation. Equality is attained when the bias and all relevant weights have magnitude $1/\sqrt{r + 1}$. We will only need the even case below.

3 Energy equality determines the output and all neuron directions

The margins imply $\mathbb{E}[\chi z] \geq 1/2$. Apply Lemma 3 separately to each neuron and then AM–GM:

$$\frac{1}{2} \leq \mathbb{E}[\chi z] \leq \rho_r \sum_j |v_j| \sqrt{b_j^2 + \|w_j\|_2^2}, \tag{6}$$

$$K_N \geq 2\sqrt{d} \sum_j |v_j| \sqrt{b_j^2 + \|w_j\|_2^2} \geq \frac{\sqrt{d}}{\rho_r}.$$

This proves the universal lower bound.

If equality holds, every nonzero neuron attains the sharp correlation bound, with its readout sign making that correlation positive. A neuron that contributes zero must have all parameters zero. Also $\mathbb{E}[\chi z] = 1/2$ and the pointwise margins force

$$z(x) = \chi(x)/2 \qquad (x \in X). \tag{7}$$

Write

$$h_s(x) = \mathrm{ReLU}\left(\frac{1}{\sqrt{r}}\sum_{i=1}^{r} s_i x_i\right), \qquad \kappa_r = \widehat{h}_{\mathbf{1}}([r]), \qquad |\kappa_r| = \rho_r.$$

Merging proportional neurons, every equality case has the form

$$z(x) = \mathrm{sign}(\kappa_r) V_* \sum_s p_s \chi(s) h_s(x), \qquad p_s \geq 0, \quad \sum_s p_s = 1, \quad V_* = \frac{1}{2\rho_r}. \tag{8}$$

Each orientation with $p_s > 0$ requires at least one nonzero neuron.

4 Fourier coefficients force exponentially many orientations

The equality-case networks ignore the other $d-r$ inputs. Throughout this section, Fourier coefficients and orientation characters are indexed by subsets of the r relevant coordinates.

Lemma 4. *Let $h = h_{\mathbf{1}}$. Its constant coefficient is positive, every even-degree coefficient is nonzero, and its odd-degree coefficients vanish except at degree one. More explicitly, if $|T| = 2t \geq 2$, $n = r-2$, and $m = 2t-2$, then*

$$\widehat{h}(T) = \frac{1}{2\sqrt{r}} \frac{\binom{n}{n/2}}{2^n} (-1)^{m/2} \frac{\binom{n/2}{m/2}}{\binom{n}{m}}. \tag{9}$$

Proof. The identity $h(x) - h(-x) = \sum_i x_i/\sqrt{r}$ gives the odd-degree claims, including $\widehat{h}(\{i\}) = 1/(2\sqrt{r})$. For $i, j \in T$, average first over these two coordinates. The remaining sum $R = \sum_{\ell \neq i,j} x_\ell$ is even, and direct evaluation gives

$$\mathbb{E}_{x_i, x_j}[x_i x_j h(x)] = \frac{1}{2\sqrt{r}} \mathbf{1}\{R = 0\}.$$

On a balanced n-bit input, $\prod_{\ell=1}^{n}(1 + u x_\ell) = (1-u^2)^{n/2}$. Comparing coefficients and using permutation symmetry shows that the conditional mean of any degree-m character on that equator is $(-1)^{m/2}\binom{n/2}{m/2}/\binom{n}{m}$. Multiplying by its probability $2^{-n}\binom{n}{n/2}$ proves (9). □

Taking Fourier coefficients in (8),

$$\widehat{z}(T) = \mathrm{sign}(\kappa_r) V_* \widehat{h}(T) \mathbb{E}_p[\chi_{[r] \triangle T}(s)]. \tag{10}$$

For every even $T \neq [r]$, the ridge coefficient is nonzero while (7) makes the left side zero. Hence

$$\mathbb{E}_p \chi_U = 0 \qquad \text{for every nonempty even } U \subseteq [r].$$

The antipodal symmetrization of p has these even Fourier moments and zero odd Fourier moments. Fourier inversion makes it uniform, so

$$p_s + p_{-s} = 2^{1-r} \qquad (s \in \{-1, 1\}^r). \tag{11}$$

Every antipodal pair therefore needs a nonzero neuron. This proves $N \geq 2^{r-1}$.

For $r = 2$, the nonzero degree-one ridge coefficients in (10) additionally force both singleton moments of p to vanish. All its nonconstant Fourier moments vanish, so p is uniform on all four orientations. Thus $N \geq 4$ in that case.

5 Matching networks and a small bounded-weight teacher

For even $r \geq 4$, let p be uniform on $s_1 = +1$. Its only nonzero Fourier moments are those of $\varnothing$ and $\{1\}$. The empty moment gives the desired parity coefficient in (10). The moment $\{1\}$ would give a coefficient at $T = [r] \setminus \{1\}$; its degree is odd and at least three, so the ridge coefficient vanishes. Therefore (8) is exactly $\chi/2$ and uses 2^{r-1} atoms. For $r = 2$, use all four orientations uniformly.

Write the nonzero atom coefficients as c_s. At any width N at least their number, set

$$(b_s, w_s) = t_s(0, s/\sqrt{r}, 0, \ldots, 0), \qquad v_s = c_s/t_s, \qquad t_s^2 = \frac{N|c_s|}{\sqrt{d}}.$$

Positive homogeneity preserves the output, and the two quadratic costs of each neuron agree. Thus the total energy is $2\sqrt{d}\sum_s |c_s| = E_*$. Zero neurons fill any additional width.

At a smaller feasible width the closed margin problem attains its minimum, because the constraints are closed and K_N is coercive. The equality characterization excludes E_*, so the minimum is strictly larger. This completes the energy and width assertions.

Finally, parity has a small bounded-weight teacher. Let $S(x) = \sum_{i=1}^{r} x_i$ and linearly interpolate the alternating values ± 1 at $S = -r, -r+2, \ldots, r$. The resulting one-variable spline has $O(r)$ hinges, initial slope of magnitude one, slope jumps of magnitude at most two, and constant term $O(r)$. Its hinge expansion gives a shallow ReLU representation of parity. Each affine form has input coefficients at most one and bias at most r. Divide it by the largest absolute coefficient and replicate the resulting readout coefficient into $O(r)$ copies, each of magnitude at most one. The constant and linear terms take $O(r)$ further neurons. All incoming and outgoing weights are then bounded by one, the total width is $O(r^2)$, and the output is exactly χ, with margin one. An ordinary XOR chain gives the $O(r)$ Boolean circuit.

6 Interpretation

The theorem exhibits an exact difference between realizability width and minimum-energy width under an unchanged isotropic Gaussian prior. It does not show that a Bayesian learner needs exponential width to learn parity, nor that an approximation to the optimal energy requires exponential width. In particular, no online input sequence or mistake lower bound is asserted. The result identifies a concrete reason why an energy calculation for a narrow network need not determine the strong-prior posterior at larger widths.

References

[1] Navid Ardeshir, Daniel Hsu, and Clayton Sanford. Intrinsic dimensionality and generalization properties of the $\mathcal{R}$-norm inductive bias. *Proceedings of the 36th Conference on Learning Theory*, PMLR 195, 2023. https://proceedings.mlr.press/v195/ardeshir23a.html.